HAICHAU HOTSPOTS

DAM501

Presented By
GROUP 2



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INTRODUCTION

1 Question

What kind of food and drink area is each part of central Da Nang ?

2 Method

Unsupervised clustering of the business mix in 250 m grid cells.

3 Result

Seven area types, delivered in an interactive 3D web map.

301,566
reviews collected

1,581
venues analysed

70
grid cells of 250 m

7
area types

11
connected zones

INTRODUCTION

What kind of food area is each neighbourhood ?

Pins show where venues are. Business mix shows what an area is like.

We describe every area by its share of coffee, rice, noodles, hotpot and other businesses.

Who it helps

1. **Venue owners**
   Find areas that suit a business concept
2. **Diners**
   See what each area is strong at
3. **Urban researchers**
   Get a structured view of the city centre

What we deliver

- Named area types
- A data profile for each type
- A web map to explore the results

INTRODUCTION

A 10-step process from problem to feedback

1

Problem definition

Area types by business mix

2

Analytic approach

Unsupervised clustering

3

Data requirements

Location, category, price, reviews

4

Data collection

Google Maps through Apify

5

Data understanding

Quality and distributions

6

Data preparation

Categories, grid and features

7

Model

K-Means, Hierarchical, GMM

8

Evaluation

Metrics, agreement, meaning

9

Deployment

Interactive web map

10

Feedback

Limits and next steps

DATA

Real reviews crawled from Google Maps with Apify

Google Maps

Public place pages and reviews

→

Apify

Google Maps Reviews Scraper actor

→

Team crawl files

Merged across the team

→

One workbook

Single source for the analysis

301,566

reviews, one per row

1,602

places

Jul 2012 – Oct 2026

range of review dates

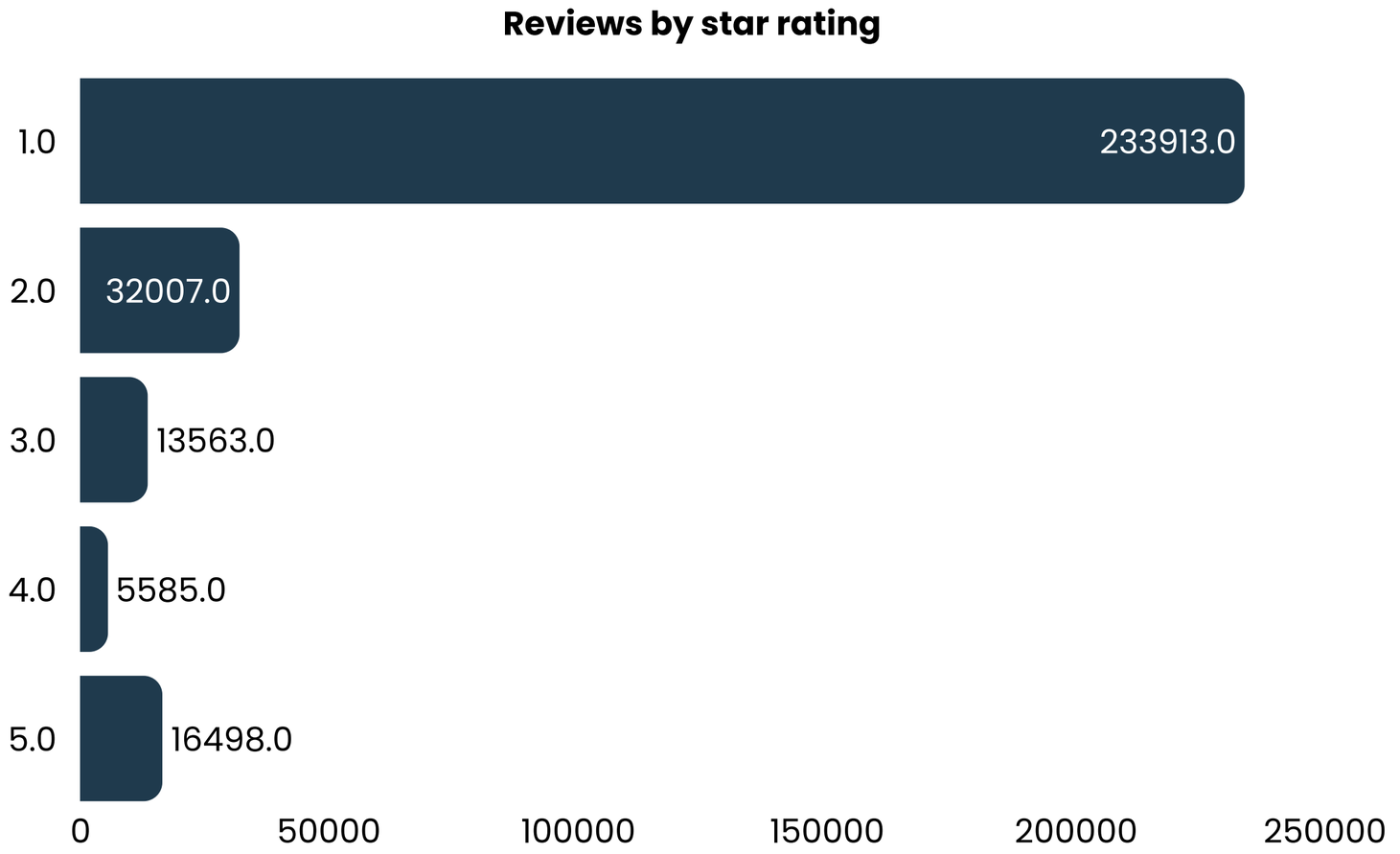
27 – 29 Oct 2026

crawl period

Each row holds the star rating, text, date, reviewer-reported price band and the details of the place.

DATA

Data understanding: ratings skew positive, coverage is uneven



78% of reviews have five stars

74% of reviews have written text

1,266 places have price information

119 places have under half of their Google reviews crawled

DATA

Cleaning: no review removed, structure made usable

Columns

161 → 62

Reviews

301,566

All kept: every review ID is unique

13,357	Wrongly parsed neighbourhoods rebuilt
2,473	English-run rows aligned with Vietnamese place details
12,318	Comment cells with HTML line breaks fixed
80,199	Missing crawl group values filled from file names
50 → 1	Image columns merged, with no data lost
15	Columns under 1% filled removed

Every rule and row count is logged in the workbook's notes sheet.

DATA

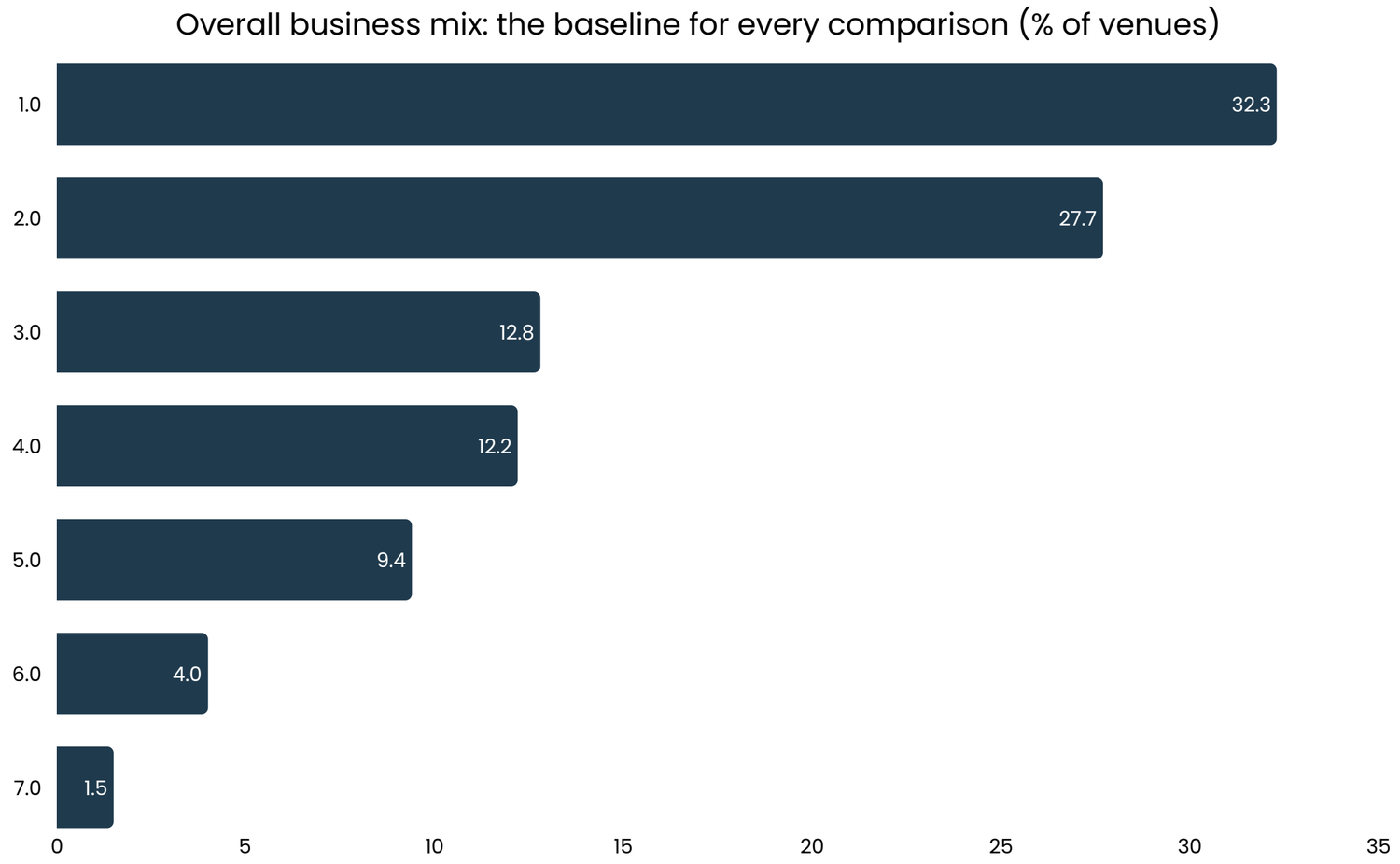
Preparation: 150 Google categories become 7 business groups

1,602 places in the data

1,600 after 2 permanently closed

1,581 after 19 non-food places

1,537 in cells with 5+ venues



MODEL AND EVALUATION

Features: each 250 m cell is described by its business mix

1 250 m grid

Equal-sized units, independent of wards

70 cells

kept from 90, with 5+ venues

2 Business mix

Seven shares per cell, smoothed with a 250 m Gaussian radius

7

features per cell

3 Hellinger

Square root of shares, suited to proportions

√p

transform

4 PCA

Keeps the main variation and reduces noise

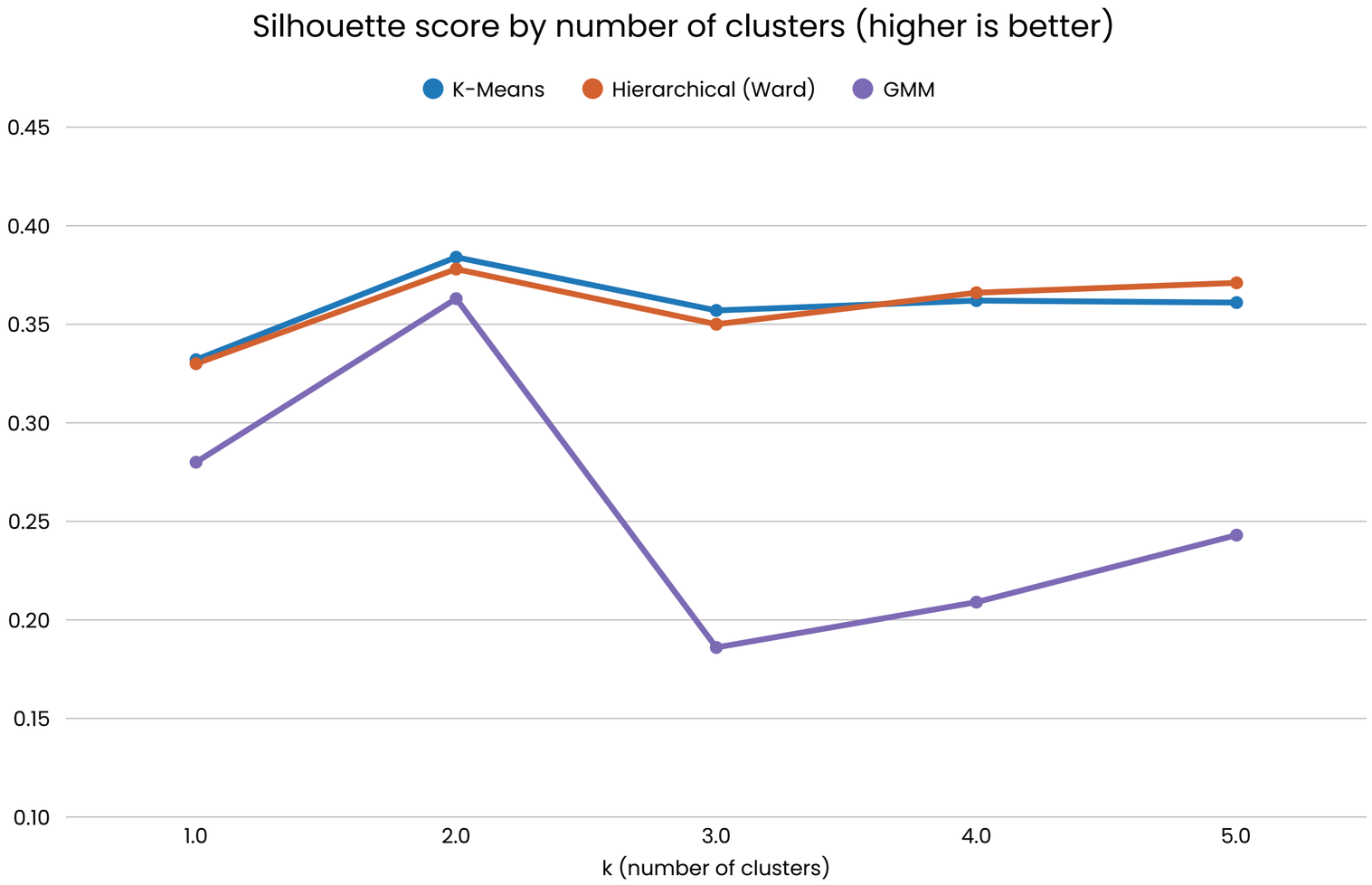
87.4%

variance in 3 components

Prices, ratings and comments are excluded from clustering. They only describe the clusters afterwards.

MODEL AND EVALUATION

Modelling: three algorithms compared across k = 3 to 7



K-Means

Centroid-based, fast and easy to explain. Selected.

Hierarchical (Ward)

Merges similar cells step by step; scores close to K-Means.

GMM

Probabilistic, but less stable: falls sharply after k = 4.

MODEL AND EVALUATION

Why k = 7: more map detail, at a measured cost

k	Silhouette	Davies–Bouldin	Calinski–Harabasz	Stability
3	0.332	1.027	36.5	0.532
4	**0.384**	0.942	43.6	**0.732**
5	0.357	0.948	43.0	0.678
6	0.362	0.893	42.8	0.602
7	0.361	**0.827**	**44.6**	0.630

K-Means. Best value per metric in orange. Lower is better for Davies–Bouldin; higher for the others.

k = 4 scores best on

Silhouette and stability, but leaves one large mixed area

k = 7 scores best on

Davies–Bouldin and Calinski–Harabasz, and gives more useful detail

Our decision

k = 7 for interpretability. Stability of 0.63 stays above our 0.5 floor

MODEL AND EVALUATION

Evaluation: moderate structure, moderate agreement

0.36
Silhouette

0.83
Davies–Bouldin

0.63
Bootstrap stability

Agreement between models at k = 7 (ARI)

	K-Means	Hierarchical	GMM
K-Means	1.00	0.58	0.57
Hierarchical	0.58	1.00	0.61
GMM	0.57	0.61	1.00

How we read this

- The main structure holds across algorithms
- Exact boundaries depend partly on the algorithm
- Moderate silhouette fits a city, where areas change gradually
- **Types are tendencies, not strict categories**

RESULT AND WEB

Result: seven area types

Area type	Cells	Venues	Avg rating	Above overall share
Hotpot, BBQ & bars + International Khu lẩu nướng · nhậu – món quốc tế	13	383	**4.54**	Hotpot, BBQ & bars ×1.6, International & healthy ×1.3
Noodles & bread + International Khu bún phở · bánh mì – món quốc tế	14	363	4.38	Noodles & bread ×1.3, International & healthy ×1.2
Coffee Khu cà phê	15	269	4.33	Drinks ×1.4
Coffee + Noodles & bread Khu cà phê – bún phở · bánh mì	9	222	4.52	Drinks ×1.3, Noodles & bread ×1.2
Snacks + Rice & Vietnamese Khu ăn vặt – cơm · món Việt	11	182	4.41	Snacks & desserts ×1.5, Rice & Vietnamese ×1.4
Rice & Vietnamese + Hotpot, BBQ & bars Khu cơm · món Việt – lẩu nướng · nhậu	4	76	4.46	Rice & Vietnamese ×1.5, Hotpot, BBQ & bars ×1.4
Hotpot, BBQ & bars + Coffee Khu lẩu nướng · nhậu – cà phê	4	42	4.35	Hotpot, BBQ & bars ×2.5, Drinks ×1.2

RESULT AND WEB

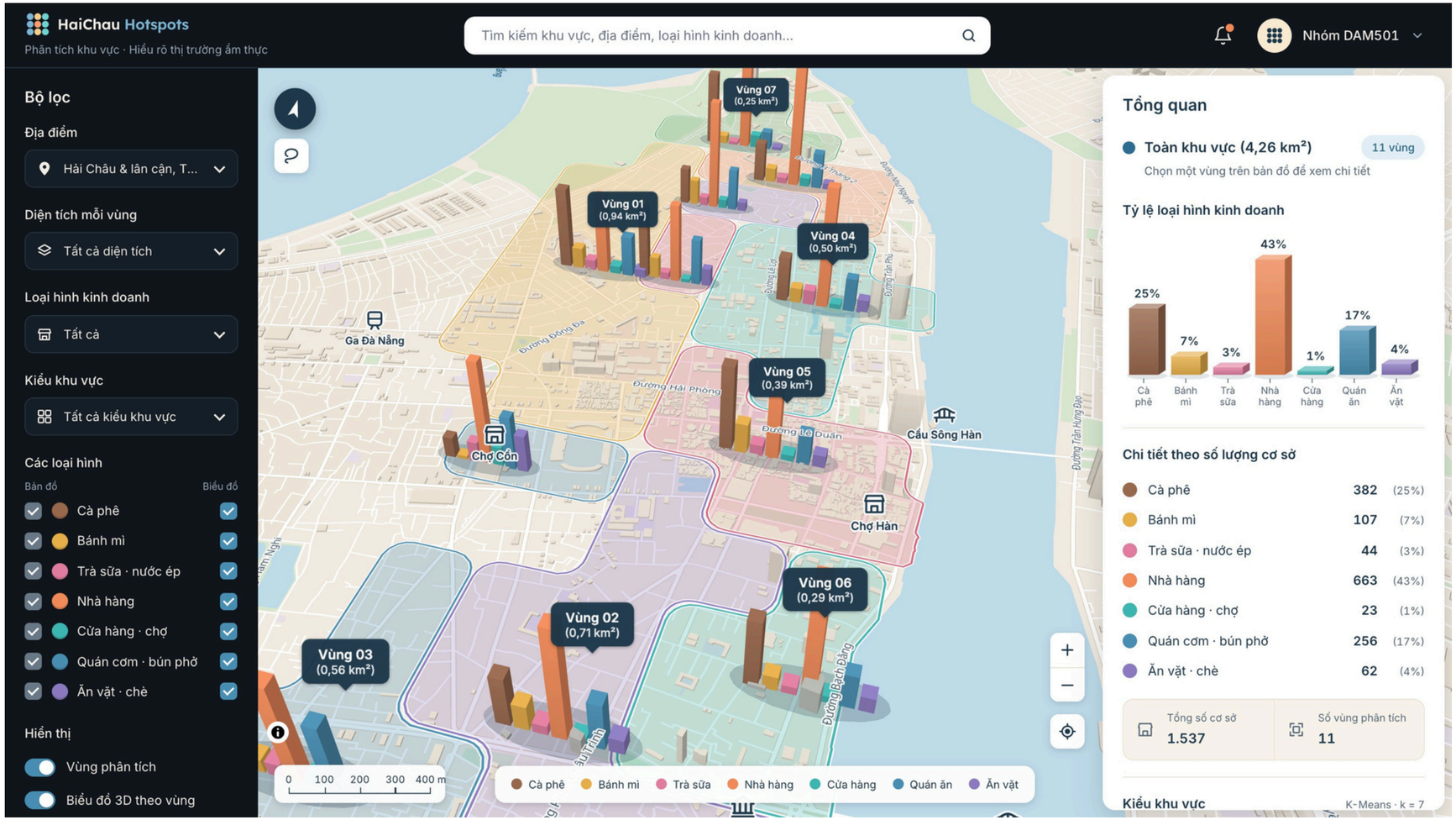
Each type is defined by what it over-indexes on

Lift by area type	Drinks	Noodles & bread	Rice & Vietnamese	Snacks	Hotpot, BBQ & bars	International	Shops & markets
Hotpot, BBQ & bars + International	0.9	0.9	0.8	*1.2*	1.6	1.3	*0.7*
Noodles & bread + International	0.8	1.3	1.1	*1.1*	0.9	1.2	*0.7*
Coffee	1.4	0.9	1.1	*0.6*	*0.3*	0.7	*1.2*
Coffee + Noodles & bread	1.3	1.2	0.8	*1.1*	0.8	0.7	*0.6*
Snacks + Rice & Vietnamese	0.8	0.8	1.4	1.5	0.6	0.9	*1.5*
Rice & Vietnamese + Hotpot, BBQ & bars	0.7	0.7	1.5	*0.3*	1.4	0.6	*1.8*
Hotpot, BBQ & bars + Coffee	1.2	*0.4*	0.7	*0.0*	2.5	0.8	*3.2*

Lift = share of a group in the type ÷ its overall share. Orange is above the overall share, teal is below. Italic values on white are groups under 5% of the type; they are not used for naming.

RESULT AND WEB

On the map, the types form 11 connected zones



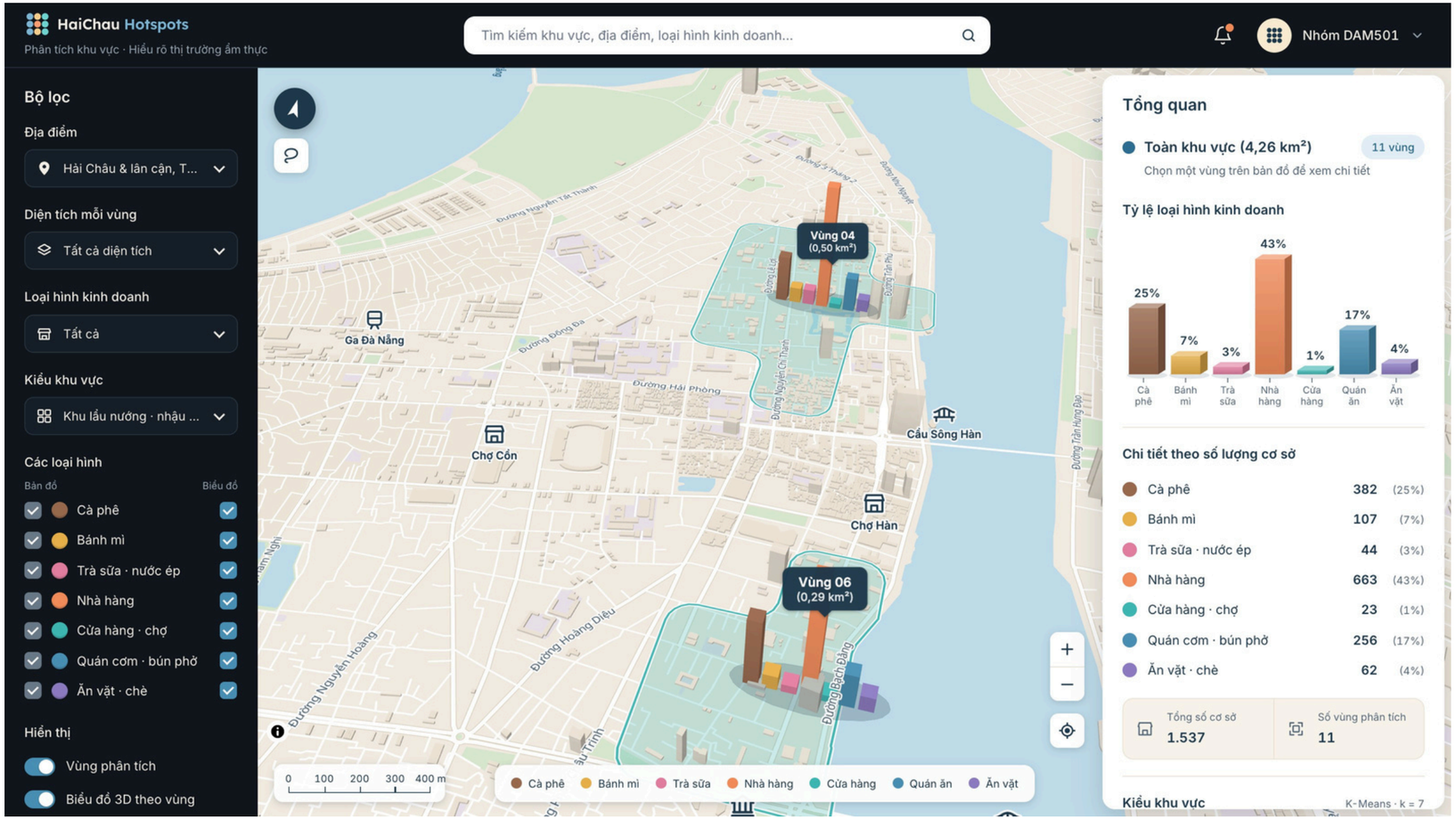
11
connected zones

4.26
km² of land covered

- Colour shows the area type
- 3D bars show the business mix
- Outlines follow the Han River

RESULT AND WEB

The web turns the results into an explorable map



Filtered to one area type. The interface is in Vietnamese for local users.

1 Filter

By place, area size, business type and area type

2 Search

Find any zone or venue

3 Draw

Circle any region for instant statistics

4 Switch

Map or satellite, light or dark

RESULT AND WEB

Inside a zone: Zone 06 explains itself with numbers

228
venues

786
venues per km², the densest zone

4.56
average Google rating

151,786
Google ratings in total

Why this name



Venues by rating



Positive comments



RESULT AND WEB

Composition separates areas; price and ratings barely do

Business mix

Works

×2.5

highest lift used in a name

Group shares differ from the overall mix by up to ×2.5, giving seven readable types.

Price

Barely separates

10 / 11

zones at up to 100,000₫

10 of 11 zones share the same most common band. The bands are wide.

Star ratings

Barely separates

80–91%

of reviews at 4+ stars per zone

Reviews are positive everywhere, so high ratings alone say little.

CONCLUSION

Limitations we report openly

1 Small sample

70 cells in the model; two types have only 4 cells

2 Untested settings

Other cell sizes and smoothing radii not yet tested

3 Uneven coverage

Favours venues on Google Maps; crawl depth varies

4 Names

Describe the business mix, not customers

5 Positive comments

Stars and keyword filters, not a sentiment model

6 Trends

Some crawl runs favoured the newest reviews

CONCLUSION

Conclusion and next steps

What we achieved

- About 300,000 raw reviews became 7 readable area types
- Results are explorable in an interactive web map
- Trade-offs and limits are reported openly

What comes next

- Test sensitivity to cell size and smoothing
- Add a real sentiment model
- Add opening hours to separate day and night areas
- Gather feedback from local users

THANK YOU

FOR YOUR ATTENTION!